



Engineering Project Management & Finance (UESTC 3031 & UESTCHN 3012)

Design For Manufacturing (Part 3)

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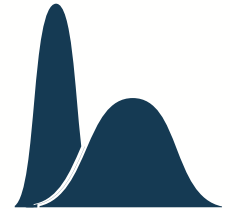
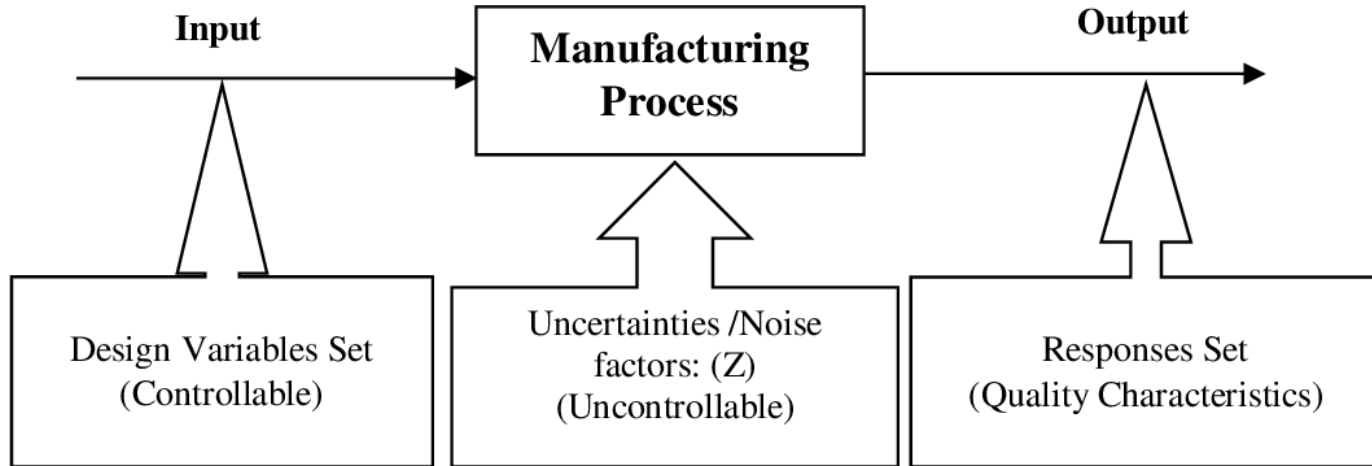
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Outline

- **Part 1:** Introduction to DFM & Design for Sustainability
- **Part 2:** Quality Control & Cost of Quality
- **Part 3:** Robust Manufacturing Design
- **Part 4:** 6-Sigma & Process Capability

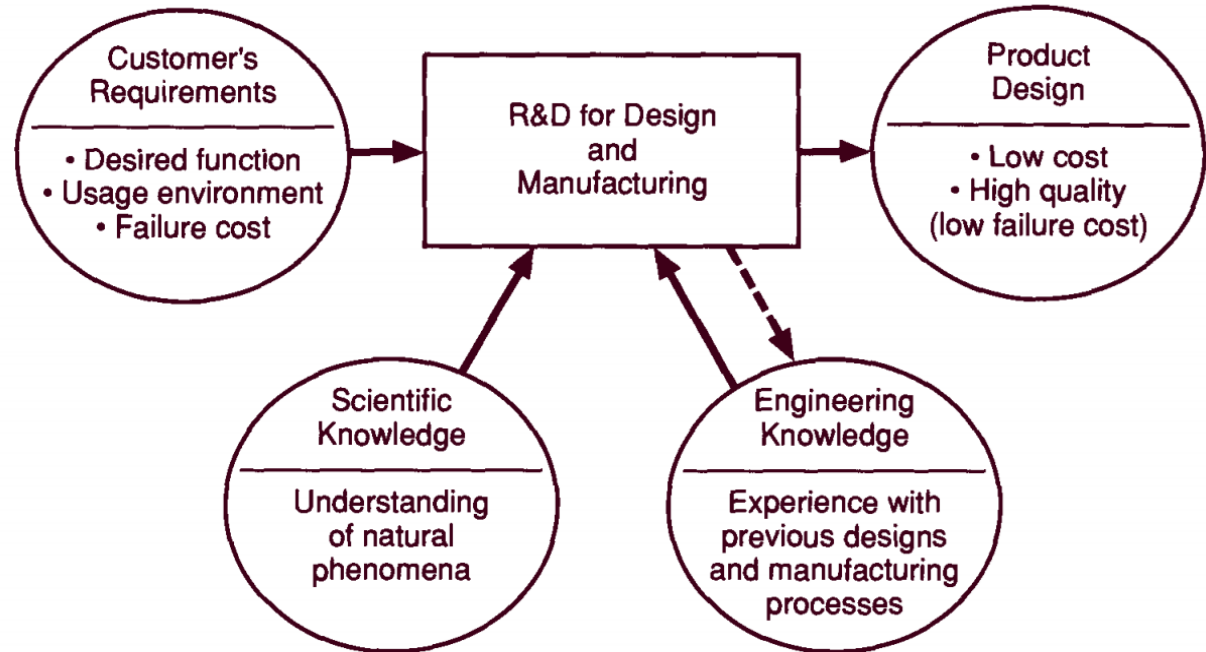


Robust Manufacturing Design



Research and Development (R&D)

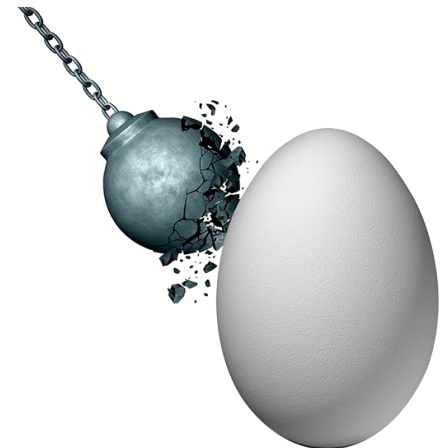
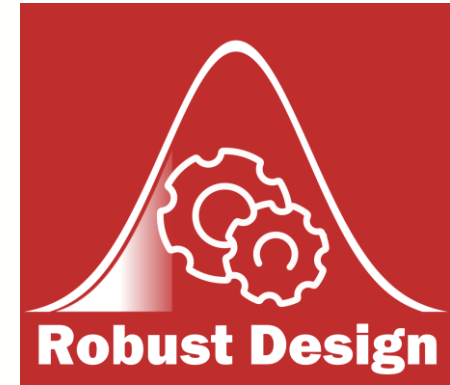
A major part of research and development (R&D), is to manufacture products that **meet customer requirements**.



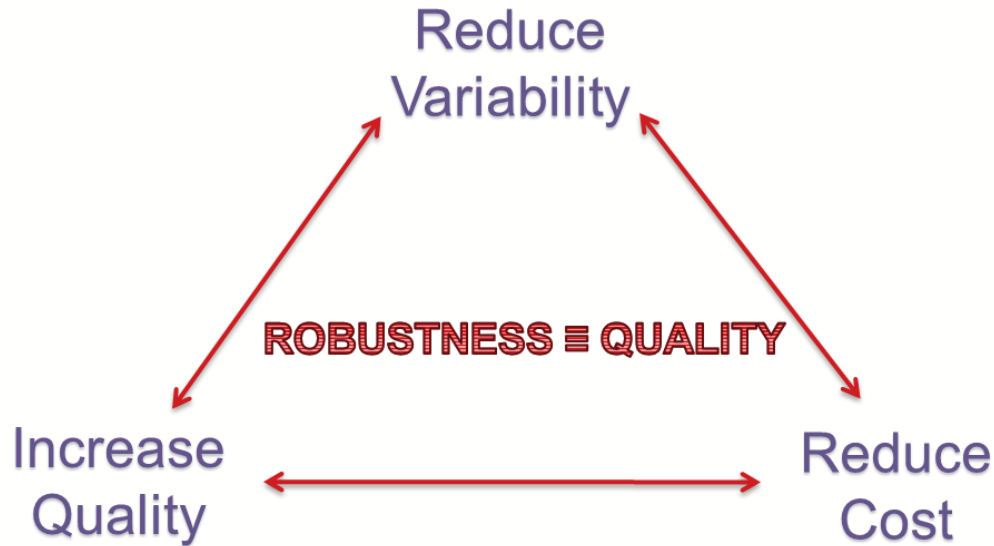
Block diagram of R&D activity.

Definition

- Robust Design is an engineering methodology used to improve productivity **during research and development (R&D)**.
- It helps produce **high-quality products** quickly and at **lower cost**.
- It uses **statistical experimental design** methods to obtain reliable information for engineering decision-making.
- The concept and foundations were developed in the 1950s–1960s by **Genichi Taguchi**.



The Basic Idea Behind Robust Design



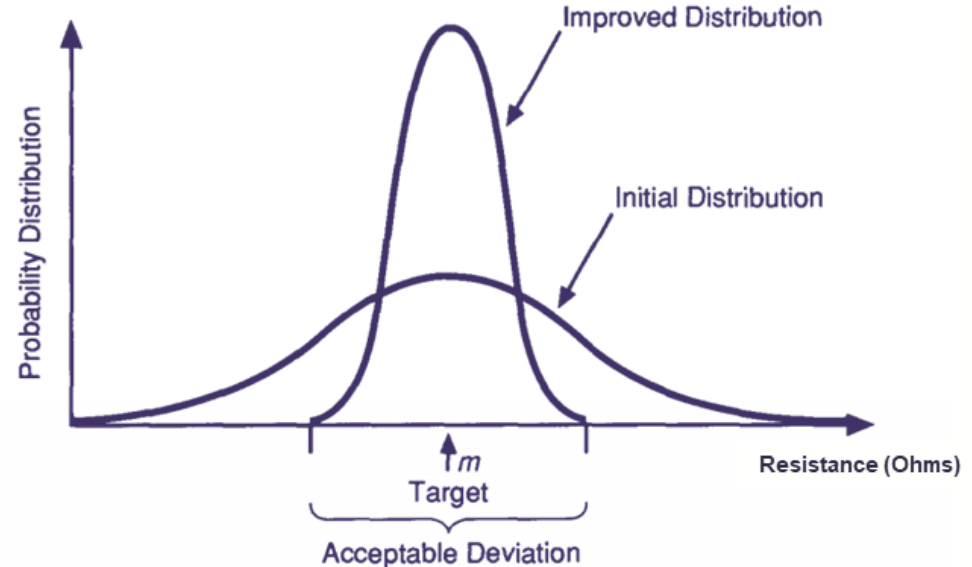
The idea behind the robust design is to **increase the quality of a process/product** by **decreasing the effects of variation** without eliminating the causes since they are too difficult or too expensive to control.

The Basic Idea Behind Robust Design

Example:

Resistor Manufacturing and Taguchi Method

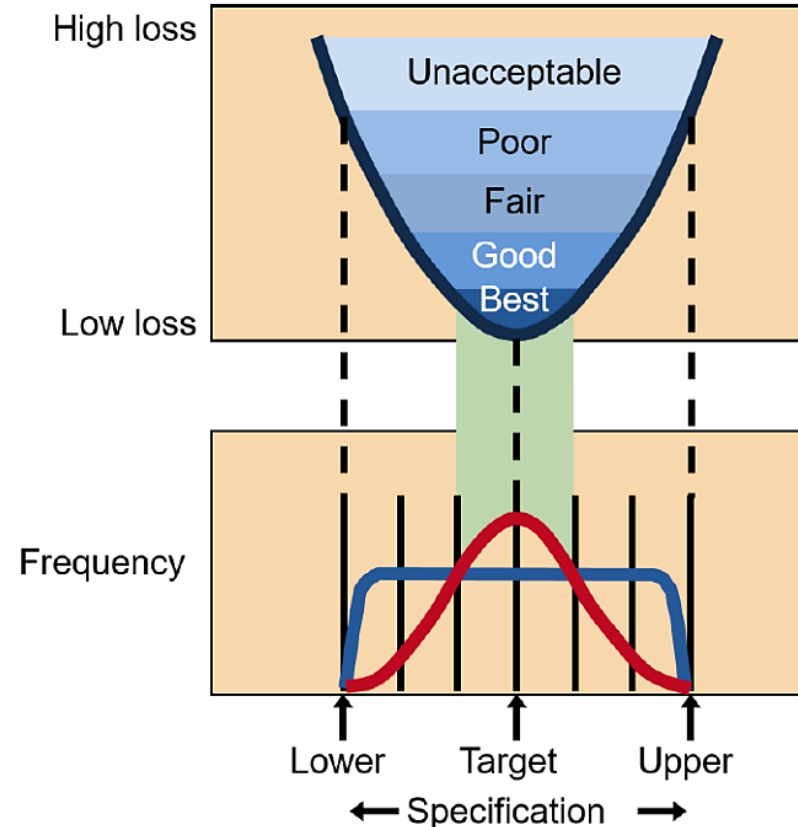
Imagine a factory produces **100-ohm resistors**, but due to variations in materials and processes, the actual resistances range from **95 ohms to 105 ohms**. This variation can affect circuit performance.



Using the Taguchi Robust Design, engineers identify and control key factors (like temperature, material quality, or machine settings) to minimize this variation. After applying Taguchi optimization, the resistors now range from **98 ohms to 102 ohms**, meaning the variation has been reduced while **still centering around 100 ohms**.

What is Quality?

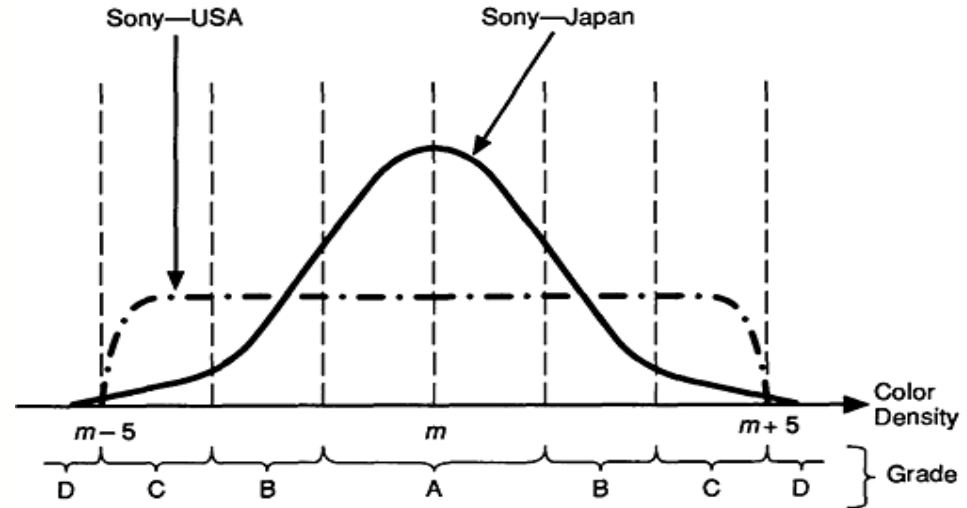
- ✓ **Ideal quality:** Customers can expect every product delivers the target performance each time the product is used (**may be impossible**).
- ✓ Following robust design (Taguchi) viewpoint, we measure the quality of a product in terms of the **total loss** to society (**producer and customer**) due to functional variation and harmful side effects.
- ✓ Under the ideal quality, the loss would be zero; the greater the loss, the lower the quality.



What is Quality?

Example 1:

In the 1970s, American consumers preferred Sony televisions made in Japan over those made in the USA, even though both used the same designs and tolerances. The reason wasn't the number of defective products but the color density. Sony-Japan's TVs had a normal distribution of color density centered around the target, with a small variation. Sony-USA's TVs had a more uniform distribution, but with wider variations. Despite Sony-USA having fewer defective units, the difference in quality was perceived due to the variation in color, not defects.

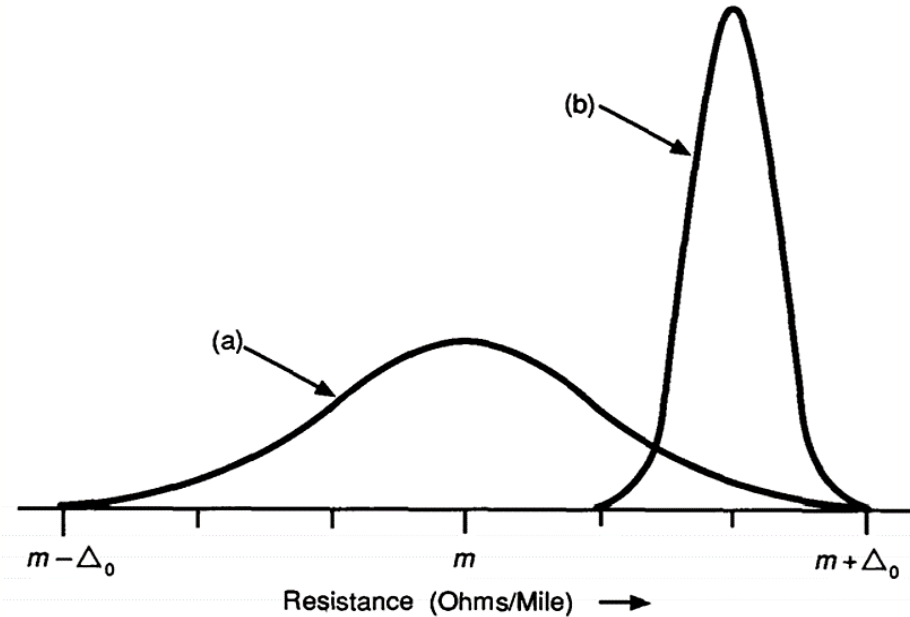


Distribution of color density in television sets. (Source: *The Asahi*, April 17, 1979).

What is Quality?

Example 2:

A manufacturer used the fraction of defective units as a quality measure for telephone cables, with a **target resistance of "m" ohms per mile**. They improved the process by reducing variance and shifting the mean resistance closer to the upper limit, reducing copper usage. However, this increase in resistance led to higher electrical losses in the network, causing complaints from users. This example demonstrates that **both variance and deviation from the target point** should be considered simultaneously for **effective process improvement**.

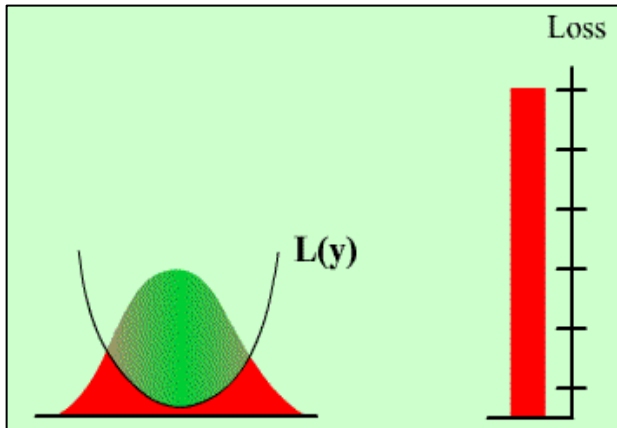


Distribution of telephone cable resistance. (a) Initial distribution. (b) After process improvement and shifting the mean.

Major Tools Used In Robust Design

The two major tools used in Robust Design are:

1. **Quality Loss Function**: measurement of quality during design/ development.
2. **Orthogonal Arrays**: efficient experimentation to find dependable information about the design parameters.

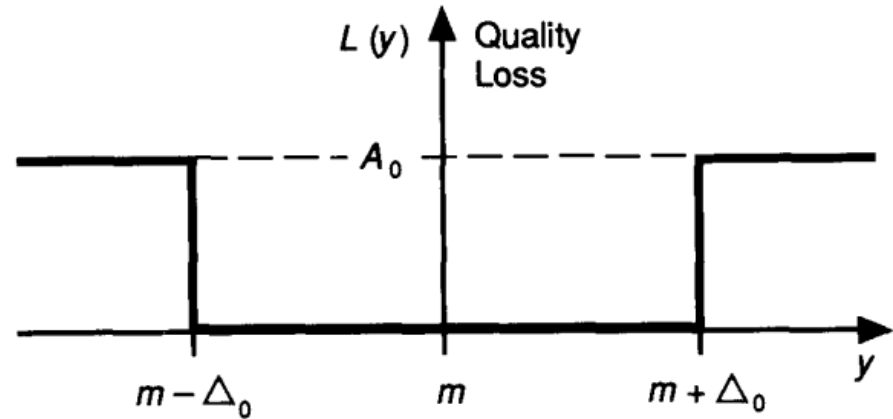


Experiment Number	Column		
	1	2	3
1	1	1	1
2	1	2	2
3	2	1	2
4	2	2	1

Traditional viewpoint

The quality loss quantify as below:

$$L(y) = \begin{cases} 0, & \text{if } |y - m| < \Delta_0 \\ A_0, & \text{otherwise} \end{cases}$$



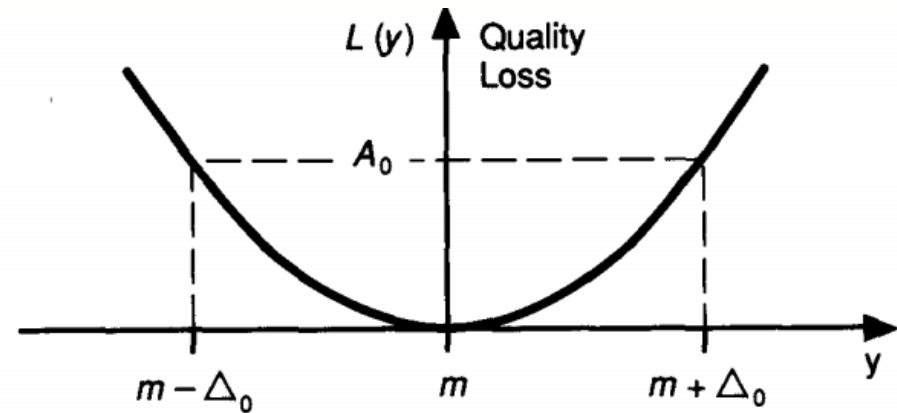
Here, A_0 is the cost of replacement or repair. Any value in the range $(m - \Delta_0)$ to $(m + \Delta_0)$ is equally good for the customer and that as soon as the range is exceeded the product is bad.

Taguchi Viewpoint (Quadratic Loss Function)

The quality loss quantify as below:

$$L(y) = k(y - m)^2$$

where k is a constant called *quality loss coefficient*:

$$k = \frac{A_0}{\Delta_0^2}$$


Notice that at $y = m$ the loss is zero. This is quite appropriate because m is the best value for y . The loss $L(y)$ increases slowly (less slope) when we are near m ; but as we go farther from m the loss increases more rapidly.

Example - Television Set Color Density:

Suppose the functional limits for the color density are $m \pm 7$. Let the repair of a television set in the field cost on average $A_0 = \$98$. The quadratic loss function can be written as:

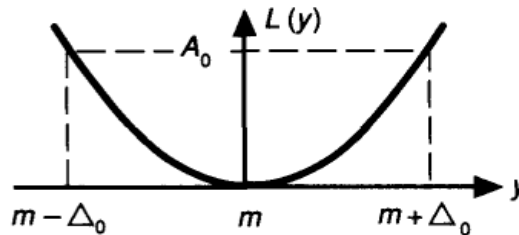
$$L(y) = \frac{98}{7^2} (y - m)^2 = 2(y - m)^2$$

Thus, the average quality loss incurred by the customers receiving sets with color density $m + 4$ is $L(m + 4) = \$32$, while customers receiving sets with color density $m + 2$ incur an average quality loss of only $L(m + 2) = \$8$.

Example - Power Supply Circuit:

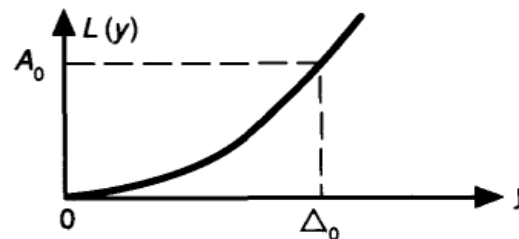
Consider a power supply circuit used in a stereo system for which the target output voltage is 110 volts. If the output voltage falls outside 110 ± 20 volts, then the stereo fails in half the situations and must be repaired. Suppose it costs \$100 to repair the stereo. Calculate the average loss associated with a particular value y of output voltage.

Types of Quality Characteristics



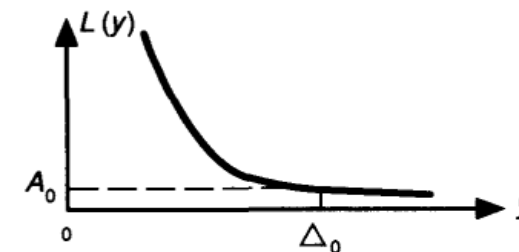
(a) Nominal-the-best

E.g., Operating Temperature, Duty Cycle, Bandwidth, Voltage Regulation, etc.



(b) Smaller-the-better

E.g., Response time of a computer, leakage current in electronic circuits, and pollution from an automobile, etc.



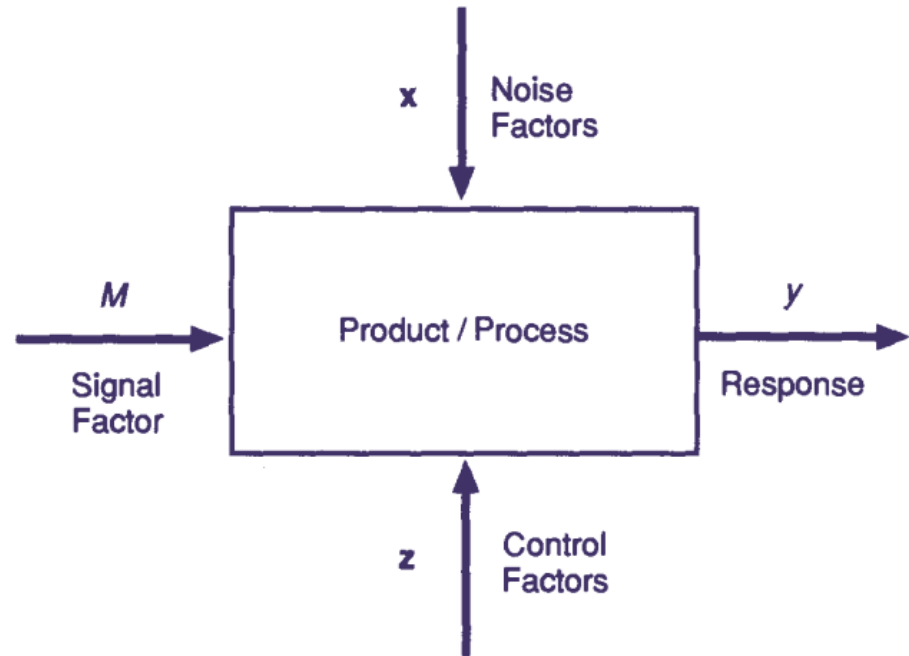
(c) Larger-the-better

E.g., Efficiency, accuracy, Power amplification, etc.

Classification of Parameters: P Diagram

- ✓ A number of parameters can influence the quality characteristic or response of the product (the word “parameter” is equivalent to the word “factor” in most of Robust Design literature):

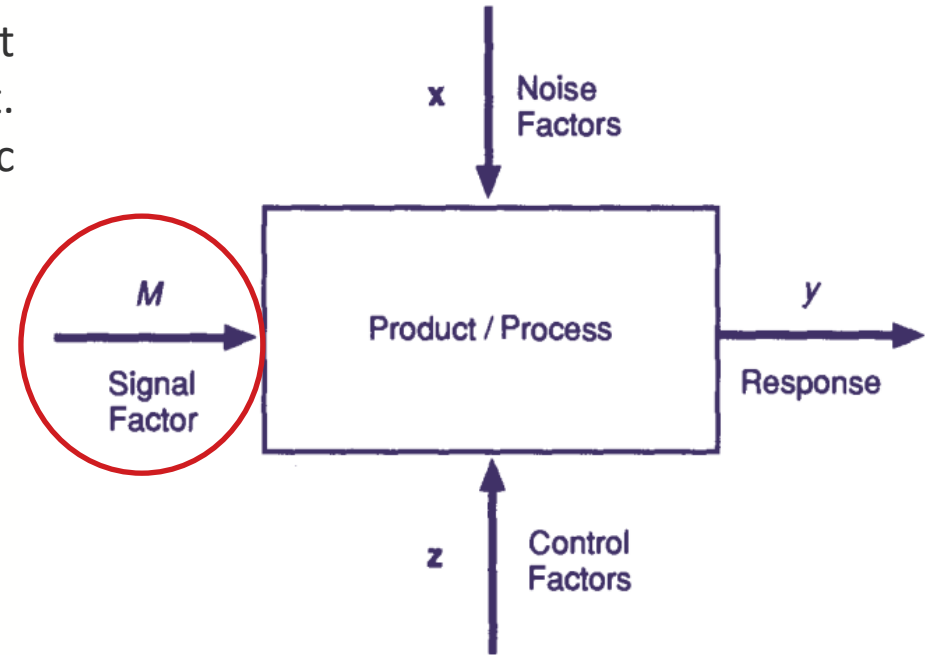
1. Signal factors
2. Noise factors
3. Control factors



Signal Factor(s)

These factors have a direct and significant impact on the performance of the product. Examples of signal factors in electronic manufacturing may include:

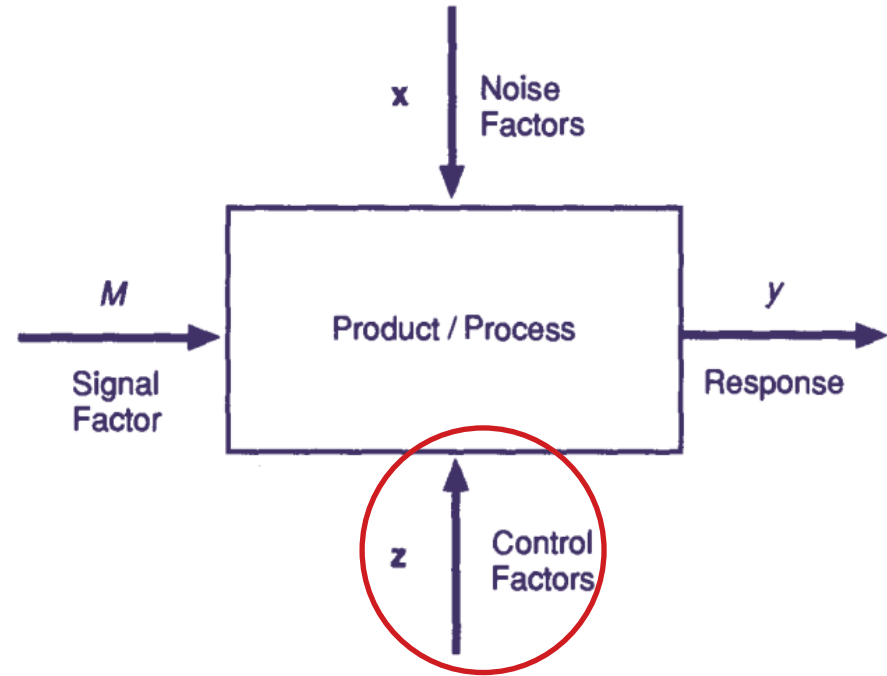
- ✓ Voltage level
- ✓ Current level
- ✓ Frequency of operation
- ✓ Resistance
- ✓ Capacitance
- ✓ Etc.



Control Factor(s)

These parameters can be specified optimally by a designer. These are equivalent to **design variables in optimization**. These factors can be controlled or manipulated to improve the performance and reliability of the product. Examples of control factors in electronic manufacturing may include:

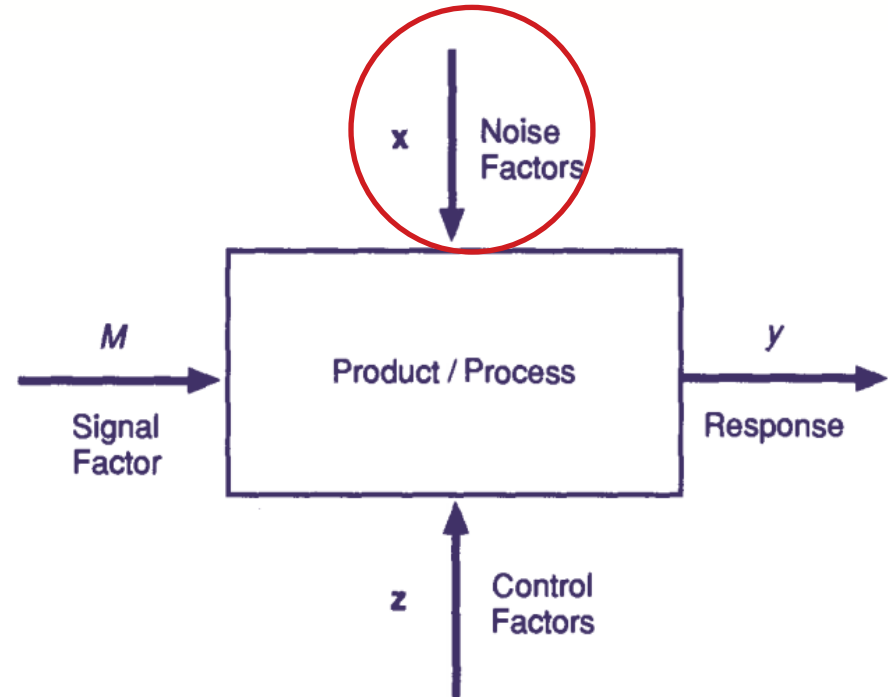
- ✓ Manufacturing process parameters
- ✓ Material properties
- ✓ Component selection
- ✓ Assembly techniques
- ✓ Testing procedures
- ✓ Etc.



Noise Factor(s)

Noise factors are factors that cause variability (uncertainty) in the performance of a system or product but cannot be controlled (or expensive) during production or product use. Examples of noise factors in electronic manufacturing may include:

- ✓ Temperature fluctuations
- ✓ Humidity levels
- ✓ Electrical interference
- ✓ Component tolerances
- ✓ Process variations
- ✓ Human error
- ✓ Etc.



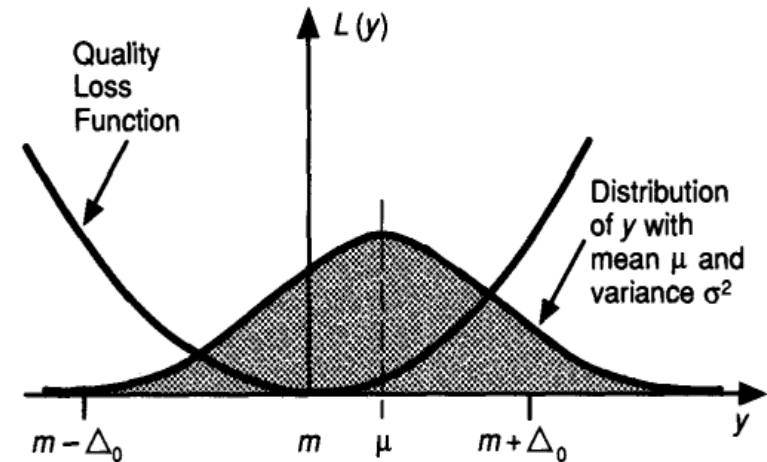
Average Quality Loss

Because of the noise factors, the quality characteristic y of a product varies from unit to unit, and from time to time during the usage of the product.

Let $y_1, y_2, \dots, y_n$ be n representative measurements of the quality characteristic y taken on a few representative units throughout the design life of the product.

Let y be a nominal-the-best type quality characteristic and m be its target value. Then, the average quality loss, Q , resulting from this product is given by:

$$Q = \frac{1}{n} [L(y_1) + L(y_2) + \dots + L(y_n)]$$

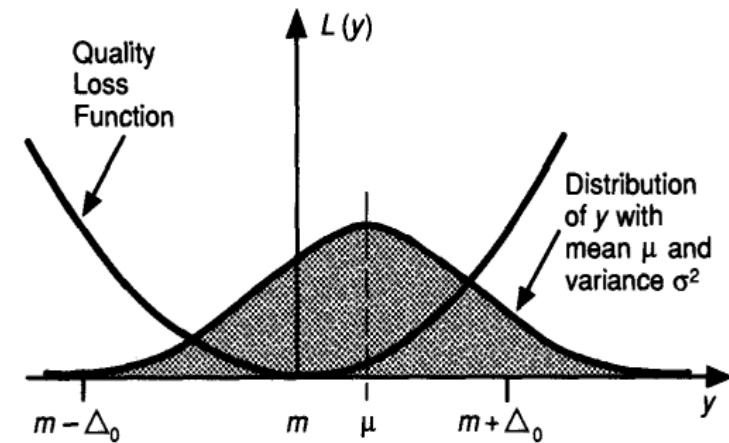


Average Quality Loss

$$Q = \frac{1}{n} [L(y_1) + L(y_2) + \cdots + L(y_n)] = \frac{k}{n} [(y_1 - m)^2 + (y_2 - m)^2 + \cdots + (y_n - m)^2]$$
$$= k \left[(\mu - m)^2 + \frac{n-1}{n} \sigma^2 \right]$$

where μ and σ^2 are the mean and the variance of y , respectively, as computed by

$$\mu = \frac{1}{n} \sum_{i=1}^n y_i \quad \sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \mu)^2$$



When n is large, then

$$Q = k[(\mu - m)^2 + \sigma^2]$$

Average Quality Loss

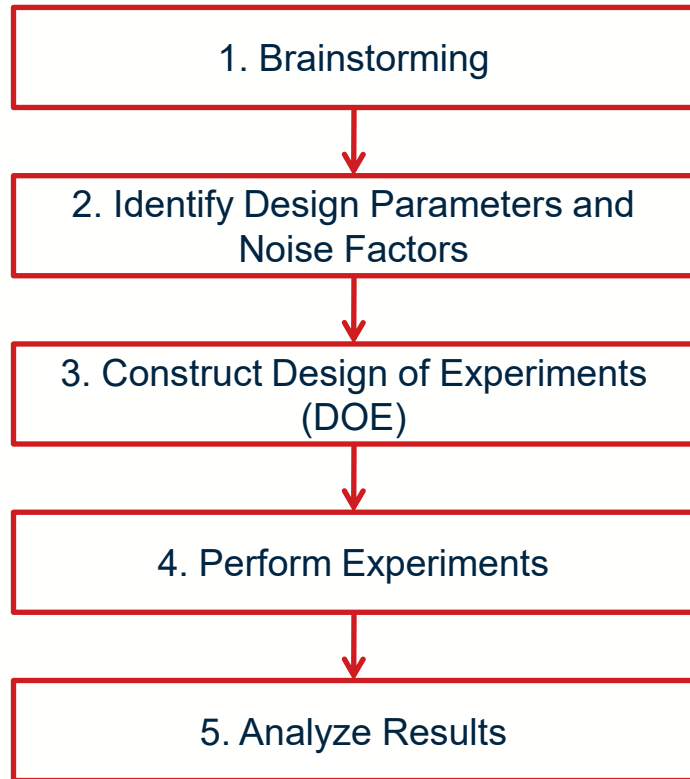
Example

Let's say that we are manufacturing a device that amplifies an input signal, and the desired output signal strength is 10 volts. However, due to various sources of variation in the manufacturing process, the output signal strength may deviate from the target value. We manufacture 10 devices, and we measure the output signal strength of each device. Here are the results:

Calculate the average quality loss function, if assume the cost of repairing of devices if fault out of 10 ± 0.5 would be \$2.

Device 1: 9.5 volts
Device 2: 9.8 volts
Device 3: 10.1 volts
Device 4: 9.9 volts
Device 5: 9.7 volts
Device 6: 10.2 volts
Device 7: 9.6 volts
Device 8: 10.3 volts
Device 9: 10.0 volts
Device 10: 9.4 volts

Taguchi Parameter Design



Design Parameters (Control Factors): Variables under your control

Noise Factors: Variables you cannot control or that are too expensive to control

Ideally, you would like to investigate all possible combinations of design parameters and noise factors and then pick the best design parameters. Unfortunately, cost and schedule constraints frequently prevent us from performing this many test cases – this is where DOE come in!

Design of Experiment (DOE)

- ✓ Full factorial always needs more experiments

Full factorial design $\rightarrow$ Number of Experiments = $(\# \text{ levels})^{\# \text{ factors}}$

Example: The total number of experiments required for 2 levels can be calculated by:

Conditions	Number of Experiments (Full factorial)
2 level 3 factors	$2^3 = 8$
2 level 4 factors	$2^4 = 16$
2 level 6 factors	$2^6 = 64$
2 level 15 factors	$2^{15} = 32,768$

- ✓ One of the common method for designing of experiments (DOE) is **Orthogonal Array (OA)**.

Design of Experiment (DOE) Orthogonal Array (OA)

Example: $L_4(2^3)$ and $L_9(3^4)$ orthogonal array

Number of
Variable Levels

Number of
Variables

$L_4(2^3)$

Number of
Experiments

$L_4(2^3)$ Orthogonal Array

Exp. Num	Variables		
	X_1	X_2	X_3
1	1	1	1
2	1	2	2
3	2	1	2
4	2	2	1

$L_9(3^4)$ Orthogonal Array

Exp. Num	Variables			
	X_1	X_2	X_3	X_4
1	1	1	1	1
2	1	2	2	2
3	1	3	3	3
4	2	1	2	3
5	2	2	3	1
6	2	3	1	2
7	3	1	3	2
8	3	2	1	3
9	3	3	2	1

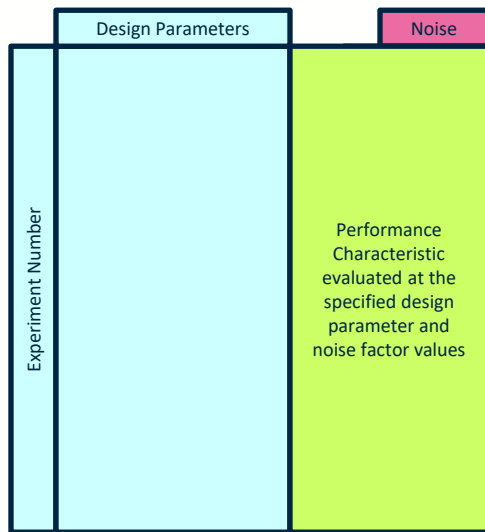
Design of Experiment (DOE)- Orthogonal Array (OA)

Standard OAs

Orthogonal Array	Number of Rows	Maximum Number of Factors	Maximum Number of Columns at These Levels			
			2	3	4	5
L_4	4	3	3	-	-	-
L_8	8	7	7	-	-	-
L_9	9	4	-	4	-	-
L_{12}	12	11	11	-	-	-
L_{16}	16	15	15	-	-	-
L'_{16}	16	5	-	-	5	-
L_{18}	18	8	1	7	-	-
L_{25}	25	6	-	-	-	6
L_{27}	27	13	-	13	-	-
L_{32}	32	31	31	-	-	-
L'_{32}	32	10	1	-	9	-
L_{36}	36	23	11	12	-	-
L'_{36}	36	16	3	13	-	-
L_{50}	50	12	1	-	-	11
L_{54}	54	26	1	25	-	-
L_{64}	64	63	63	-	-	-
L'_{64}	64	21	-	-	21	-
L_{81}	81	40	-	40	-	-

Design of Experiment (DOE)- Orthogonal Array (OA)

Inner & Outer Arrays (Crossed Arrays)

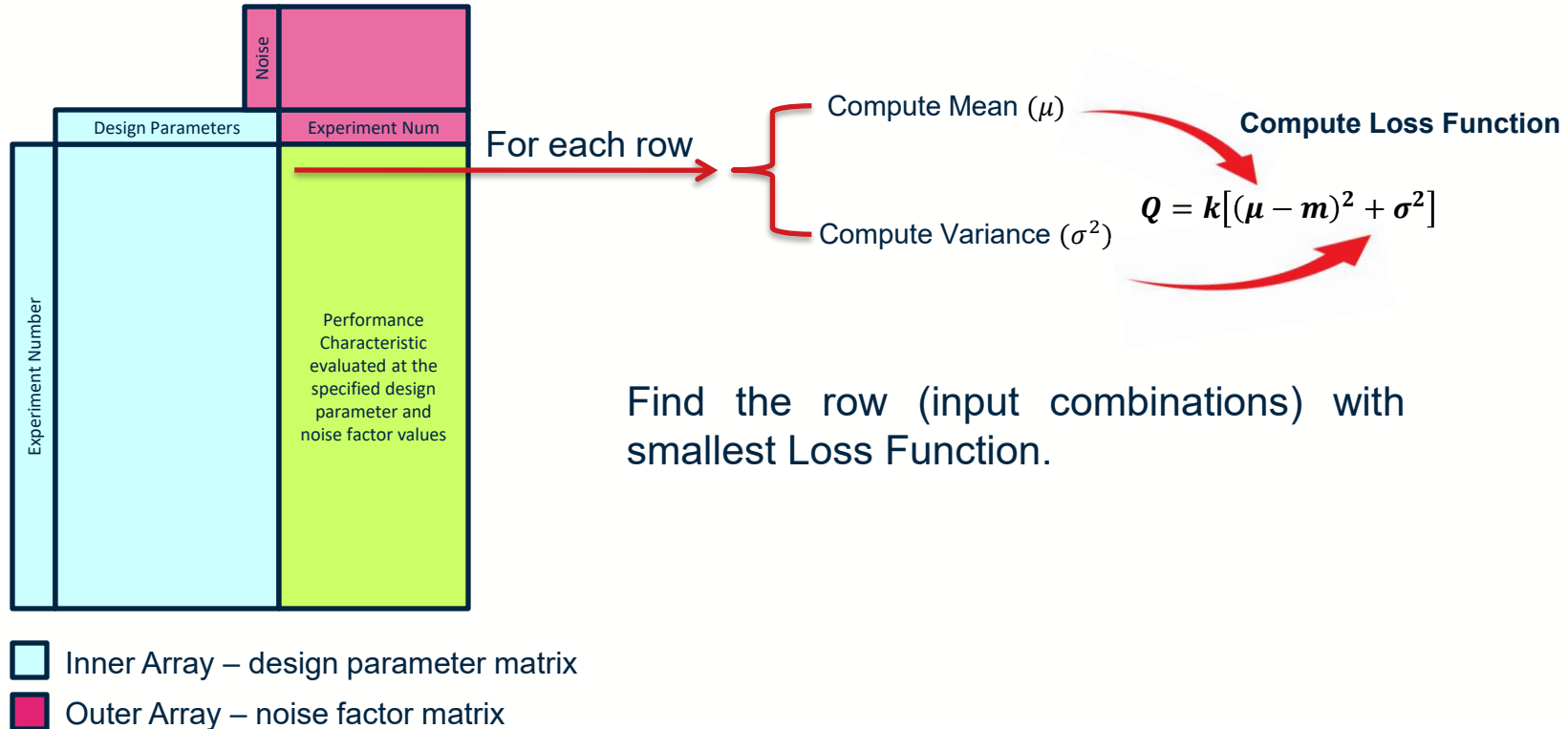


	X_1	X_2	X_3	X_4	N_3	N_2	N_1
1	1	1	1	1	1	1	1
2	1	2	2	2	1	2	1
3	1	3	3	3	1	1	2
4	2	1	2	3	2	2	2
5	2	2	3	1	2	1	1
6	2	3	1	2	1	2	2
7	3	1	3	2	2	1	1
8	3	2	1	3	1	2	2
9	3	3	2	1	2	1	1

$y_{11} = f\{X_1(1), X_2(1), X_3(1), X_4(1), N_1(1), N_2(1), N_3(1)\}$
 $y_{52} = f\{X_1(2), X_2(2), X_3(3), X_4(1), N_1(1), N_2(2), N_3(2)\}$

- Inner Array – design parameter matrix
- Outer Array – noise factor matrix

Design of Experiment (DOE)- Orthogonal Array (OA)



Example

There are four controllable factors (A, B, C, D) each at three levels and three noise or uncontrollable factors (E, F, G) each at two levels. Using the experimental results in this crossed arrays design, find the relevant robust optimal product design (assume the target point for the product is 20, and $K=1$ in quality loss function).

(b) Outer Array												
					<i>E</i>	–	–	–	–	+	+	+
					<i>F</i>	–	–	+	+	–	–	+
					<i>G</i>	–	+	–	+	–	–	+

(a) Inner Array												
Run	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>								
1	–1	–1	–1	–1	15.6	9.5	16.9	19.9	19.6	19.6	20.0	19.1
2	–1	0	0	0	15.0	16.2	19.4	19.2	19.7	19.8	24.2	21.9
3	–1	+1	+1	+1	16.3	16.7	19.1	15.6	22.6	18.2	23.3	20.4
4	0	–1	0	+1	18.3	17.4	18.9	18.6	21.0	18.9	23.2	24.7
5	0	0	+1	–1	19.7	18.6	19.4	25.1	25.6	21.4	27.5	25.3
6	0	+1	–1	0	16.2	16.3	20.0	19.8	14.7	19.6	22.5	24.7
7	+1	–1	+1	0	16.4	19.1	18.4	23.6	16.8	18.6	24.3	21.6
8	+1	0	–1	+1	14.2	15.6	15.1	16.8	17.8	19.6	23.2	24.2
9	+1	+1	0	–1	16.1	19.9	19.3	17.3	23.1	22.7	22.6	28.6





According to the Taguchi quality philosophy, the quadratic loss function can be represented in several forms depending on the design objective. Which of the following is NOT one of the common Taguchi loss function types?

- A. Nominal-the-best
- B. Smaller-the-better
- C. Larger-the-better
- D. Target-the-zero
- E. None of the above



In robust design, the response of a product (denoted by y) can be influenced by three classes of parameters represented as Z , M , and X . Which of the following options correctly represents two of these parameter classes?

- A. Control factors and Noise factors
- B. Marketing factors and Economic factors
- C. Environmental factors and Financial factors
- D. Human factors and Cost factors
- E. Demand factors and Supply factors



A product engineer collects 10 observations for a product quality characteristic where the objective is nominal-the-best. The target value is 8.5 and the Taguchi loss constant $k=1$. The observations are:

8.100, 8.900, 8.450, 9.250, 8.860, 8.350, 8.250, 8.680, 8.900, 9.050

Based on these data (nominal the best), what is the average Taguchi loss value for this dataset?

- A. 0.3762
- B. 0.1736
- C. 0.0319
- D. 0.6790
- E. 0.1420

In the Next Lesson:

Part 4: 6-Sigma & Process Capability



Appendix: Taguchi OA samples

- L4: Three two-level factors
- L8: Seven two-level factors
- L9: Four three-level factors
- L12: Eleven two-level factors
- L16: Fifteen two-level factors
- L16b: Five four-level factors
- L18: One two-level and seven three-level factors
- L25: Six five-level factors
- L27: Thirteen three-level factors
- L32: Thirty-two two-level factors
- L32b: One two-level factor and nine four-level factors
- L36: Eleven two-level factors and twelve three-level factors
- L50: One two-level factors at 2 levels and eleven five-level factors
- L54: One two-level factor and twenty-five three-level factors
- L64: Thirty-one two-level factors
- L64b: Twenty-one four-level factors
- L81: Forty three-level factors

